Foundations of Full House Methodology

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Abstract

The Full House Methodology is an approach for computing era-adjusted or system-adjusted statistics that balances the size of the eligible population in each system with the component's performance via "versus their peers" in a specific system. We assume that each component has a latent aptitude that can be calculated from this balancing act provided the system inclusion mechanism for components of the system is known. Without making any assumptions about the distribution of the system's components, the components' distribution is well-established from the nonparametric probability distribution.

1 Introduction

Several academics have attempted to develop methods that can correct for era or system biases. The starting place for the answer to the problem of comparing players across eras can be found in the work of Stephen Jay Gould. Gould [1996] suggested that the entire distribution of achievement, or "full house of variation," is relevant when considering the achievements of baseball players across eras. This idea was also recognized as the starting plac e for an answer by Schell [2013] and Schell [2016].

Berry et al. [1999], Petersen et al. [2011], Schell [2013, 2016], Petersen and Penner [2020] provide several techniques to compare baseball players across eras, but they have fallen short in their objective and their results remain biased, as discussed in Eck [2020a] and Eck [2020b]. In short, these previous attempts do not incorporate the evolving talent pool in calculations, which is necessary for understanding the changing distributions of achievement over time. The result is that existing methodologies produce a disproportionate number of players who began their career before baseball was integrated players in the rankings of baseball's greatest of all time. This era of baseball did not include a crucial population of eligible players. Yu et al. [2016], Meyer [2021], Campbell [1990] also develop methods for comparing components in historical popularity, educational attainment, and evaluation of employees. But they have similar problems as stated before and fail to account for the evolving talent pool.

In some circumstances, such as test equating, it is not required to take into account an evolving talent pool. Andersson et al. [2013] offers a collaborative framework for test equating and builds the cumulative density function using the kernel method. Nevertheless,

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they are unable to resolve the extrapolation issue in non-parametric estimation, and the distribution's tail is not given special consideration.

Our Full House methodology directly incorporates the quality of the talent pool as a central modeling component. It offers metrics for assessing the performance of components in various eras or systems. With the application of educational attainment simulation, our Full House methodology reveals that the most talented students could perform worse than the other students who attend schools with lower mean scores.

The next section establishes the model setting and specifies the assumptions in our Full House methodology. Section 3 considers an intriguing paradox that the component with more talent underperforms the component with less talent. We compute the probability of this paradox under different scenarios. Section 4 discusses the application of Full House methodology for the educational attainment simulation.

2 Model Setting

We now motivate the structure of the Full House Model. Let N_i denote the size of population i. We will suppose the every individual $k = 1, \ldots, N_i$ has an underlying latent aptitude $X_{i,1}, \ldots, X_{i,N_i} \overset{\text{iid}}{\sim} F_X$. We denote $X_{i,(j)}$ as the jth ordered latent aptitude in population i. We define $g_i(\cdot, \cdot)$ as the system inclusion function, where $g_i(X_{i,j}, \mathbf{X}_{i,-j}) = 1$ indicates that individual j is included in the ith system, and $g_i(X_{i,j}, \mathbf{X}_{i,-j}) = 0$ indicates that individual j is not included in the ith system. Let $\mathbf{X}_{i,-j}$ be the vector of all latent aptitudes not including the component j.

We now suppose that an individual j included in system i has an observed statistic $Y_{i,j}$ where we suppose that $Y_{i,1}, \ldots, Y_{i,n_i} \overset{iid}{\sim} F_{Y_i}$, F_{Y_i} is continuous, and $n_i < N_i$ is the number of individuals included in system i. We denote $Y_{i,(j)}$ as the jth ordered statistic for system i.

The key to the Full House Model is connecting latent aptitudes X with observed statistics Y. To achieve this we assume that we can form pairs $(Y_{i,(j)}, X_{i,(r(N_i,n_i,j))})$, $j = 1, \ldots, n_i$, where $r(N_i, n_i, k)$ is the index function that connects this latent aptitude to the kth components in the system i. The function $r(N_i, n_i, k)$ determines the relationship between the components in the system and the selected latent aptitudes in the population. As an example, $r(N_i, n_i, k) = N_i - n_i + k$, then

We now demonstrate how the Full House Model works in a parametric setting. Consider the pair $(Y_{i,(j)}, X_{i,(r(N_i,n_i,j))})$ and suppose that the distribution corresponding to $Y_{i,j}$ from the ith system is known to belong to a continuous parametric family indexed by unknown parameter θ_i, and let $F_{Y_i}(\cdot \mid \theta_i)$ be a parametric CDF with parameter $\theta_i \in \mathbb{R}^{p_{Y_i}}$. We can estimate θ_i with $\hat{\theta}_i$ and plug the estimator into the CDF $F_Y(\cdot \mid \hat{\theta}_i)$.

In order to connect the $Y_{i,(j)}$ and $X_{i,(r(N_i,n_i,j))}$ and obtain an estimate of the underlying talents, we will make use of the following classical order statistics properties,

$$F_{Y_i}\left(Y_{i,(j)} \mid \theta_i\right) \sim U_{i,(j)}, \quad F_{Y_i}\left(Y_{i,(j)} \mid \hat{\theta}_i\right) \approx U_{i,(j)},$$
$$F_{Y_{i,(j)}}\left(Y_{i,(j)} \mid \theta_i\right) \sim U_{i,j}, \quad F_{Y_{i,(j)}}\left(Y_{i,(j)} \mid \hat{\theta}_i\right) \approx U_{i,j},$$

where $\approx$ means approximately distributed, $\sim$ means distributed as, $U_{i,j} \sim U(0,1)$, and $U_{i,(j)} \sim \text{Beta}(j, n_i + 1 - j)$ and the quality of the approximation in the right hand side depends upon the estimator $\hat{\theta}_i$ and the sample size.

We now connect the order statistics to the underlying talent distribution that comes from a population with $N_i \geq n_i$ observations when F_X is known. This connection is established with

$$F^{-1}_{X_{i,(r(N_i,n_i,j))}} \left(F_{Y_{i,(j)}} \left(Y_{i,(j)} \mid \theta_i \right) \right) \sim F^{-1}_{X_{i,(r(N_i,n_i,j))}} \left(U_{i,j} \right) = X_{i,(r(N_i,n_i,j))}. \tag{1}$$

We estimate the above with

$$F^{-1}_{X_{i,(r(N_i,n_i,j))}} \left(F_{Y_{i,(j)}} \left(Y_{i,(j)} \mid \hat{\theta}_i \right) \right) \approx F^{-1}_{X_{i,(r(N_i,n_i,j))}} \left(U_{i,j} \right) = \widehat{X}_{i,(r(N_i,n_i,j))}. \tag{2}$$

Similarly, we can obtain Y from X,

$$F^{-1}_{Y_{i,(j)}} \left(F_{X_{i,(r(N_i,n_i,j))}} \left(X_{i,(r(N_i,n_i,j))} \right) | \theta_i \right) = Y_{i,(j)}, \tag{3}$$

where we estimate the above with

$$F^{-1}_{Y_{i,(j)}} \left(F_{X_{i,(r(N_i,n_i,j))}} \left(X_{i,(r(N_i,n_i,j))} \right) | \hat{\theta}_i \right) = \widehat{Y}_{i,(j)}.$$

Not sure where cross-contextual comparisons are needed. We can use the Full House Model to obtain estimated statistics in another context. Consider the the pair $(Y_{i,(j)}, X_{i,(r(N_i,n_i,j))})$ in the ith population. We first put the aptitude value $X_{i,(r(N_i,n_i,j))}$ obtained by (2) in a new population m and reverse the process to obtain the hypothetical statistics in system m, which we will denote as $Y_{i,j,m}$. More formally, $Y_{i,j,m}$ are estimated as follows in the parametric context,

$$\widehat{Y}_{i,j,m} = F^{-1}_{Y_m} \left(F_{X_{m,(r(N_m,n_m,l_{i,j,m}))}} \left(\widehat{X}_{i,(r(N_i,n_i,j))} \right) | \hat{\theta}_m \right), \tag{4}$$

where $l_{i,j,m}$ is the rank of $\widehat{X}_{i,(r(N_i,n_i,j))}$ among the values

$$\left\{ \widehat{X}_{i,(r(N_i,n_i,j))}, \widehat{X}_{m,(r(N_m,n_m,t))} : t = 1, \ldots, n_m \right\}.$$

The estimates (4) correspond to

$$Y_{i,j,m} = F^{-1}_{Y_m} \left(F_{X_{m,(r(N_m,n_m,l_{i,j,m}))}} \left(X_{i,(r(N_i,n_i,j))} \right) \mid \theta_m \right), \tag{5}$$

where $l_{i,j,m}$ is the rank of $X_{i,(r(N_i,n_i,j))}$ among the values

$$\left\{ X_{i,(r(N_i,n_i,j))}, X_{m,(r(N_m,n_m,t))} : t = 1, \ldots, n_m \right\},$$

and $X_{i,(r(N_i,n_i,j))}$ is computed via (1).

3 Paradoxes

Consider two populations $i = 1, 2$ with sizes N_1 and N_2, and systems of sizes n_1 and n_2. We define a paradox (or a Gould paradox) when

$$X_{1,(N_1)} > X_{2,(N_2)} \quad \text{and} \quad Y_{1,(n_1)} < Y_{2,(n_2)}$$

both occur. A central interest is evaluating the probability of such a paradox,

$$\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right). \tag{6}$$

As a simple illustrative example of some mechanics concerning paradoxes, suppose that $Y_{1,j} < Y_{2,k}$ a.s. Then (6) is equivalent to $\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}\right)$, and we have $\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}\right) \to 1$ as $N_1 \to \infty$. We motivate a convenient representation of (6) in the following proposition.

Proposition 3.1. *Consider two populations $i = 1, 2$ with sizes N_1 and N_2, and systems of sizes n_1 and n_2. Suppose that the system inclusion mechanism has the property that $g(X_{i,(N_i)}) = 1$, and that the Full House Model is of the form*

$$\begin{aligned} Y_{i,j} &\sim F_Y(y), \qquad i = 1, 2, \qquad j = 1, \ldots, n_1, \\ X_{i,j} &\sim F_X(x), \qquad i = 1, 2, \qquad j = 1, \ldots, N_i, \end{aligned}$$

where $F_Y^{-1}(y)$ and $F_X^{-1}(x)$ exist. Then,

$$\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) = \mathrm{P}\left(X_{2,(N_2)} < X_{1,(N_1)} < h_{1,2}\left(X_{2,(N_2)}\right)\right), \tag{7}$$

where $h_{i,j}(\bullet) = (f_i^{-1} \circ f_j)(\bullet)$, $f_i(\bullet) = F^{-1}_{Y_{i,(n_i)}}\left(F_{X_{i,(N_i)}}(\bullet)\right)$, and both f and h exist.

Proof. If f and h exist, then we have

$$\begin{aligned} &\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) \\ &\quad = \mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, f_1(X_{1,(N_1)}) < f_2(X_{2,(N_2)})\right) \\ &\quad = \mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, f_1(X_{1,(N_1)}) < f_2(X_{2,(N_2)})\right) \\ &\quad = \mathrm{P}\left(X_{2,(N_2)} < X_{1,(N_1)} < h_{1,2}\left(X_{2,(N_2)}\right)\right). \end{aligned}$$

Existance of f_i follows directly from the existance of $F_Y^{-1}(y)$, and existance of $h_{i,j}$ follows directly from the existance of f_i and the existance of $F_X^{-1}(x)$. □

Lemma 3.2. *Suppose that the system inclusion mechanism has the property that $g(X_{i,(N_i)}) = 1$, and that the Full House Model is of the form*

$$\begin{aligned} Y_{i,j} &\sim F_Y(y), \qquad i = 1, 2, \qquad j = 1, \ldots, n_1, \\ X_{i,j} &\sim F_X(x), \qquad i = 1, 2, \qquad j = 1, \ldots, N_i, \end{aligned}$$

where $F_Y^{-1}(y)$ and $F_X^{-1}(x)$ exist. Then,

$$\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) = \frac{n_2}{n_1 + n_2} - \frac{N_2}{N_1 + N_2}.$$

Proof. We can use the convenient representation of (7) since the conditions of Proposition 3.1 hold. Thus, we have

$$
\begin{aligned}
&\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) = \mathrm{P}\left(X_{2,(N_2)} < X_{1,(N_1)} < h_{1,2}\left(X_{2,(N_2)}\right)\right) \\
&= \int_{-\infty}^{+\infty} \left[\int_{x}^{h_{1,2}(x)} f_{X_{1,(N_1)}}(y)\, dy \right] f_{X_{2,(N_2)}}(x)\, dx \\
&= \int_{-\infty}^{+\infty} \left[\int_{x}^{h_{1,2}(x)} N_1 (F_X(y))^{N_1-1} f_X(y) dy \right] N_2 (F_X(x))^{N_2-1} f_X(x)\, dx \\
&= \int_{-\infty}^{+\infty} \left[\int_{x}^{F_X^{-1}(F_X(x)^{\frac{N_2}{N_1}\frac{n_1}{n_2}})} N_1 (F_X(y))^{N_1-1} f_X(y) dy \right] N_2 (F_X(x))^{N_2-1} f_X(x)\, dx \\
&= \int_{-\infty}^{+\infty} \left(F_X(x)^{N_2\frac{n_1}{n_2}} - F_X(x)^{N_1}\right) N_2 (F_X(x))^{N_2-1} f_X(x)\, dx \\
&= \int_{0}^{1} t^{\frac{n_1}{n_2}} - t^{\frac{N_1}{N_2}}\, dt \\
&= \frac{n_2}{n_1+n_2} - \frac{N_2}{N_1+N_2},
\end{aligned}
$$

where

$$
h_{1,2}(x) = F_X^{-1}((F_Y(F_Y^{-1}(F_X^{\frac{N_2}{n_2}})))^{\frac{n_1}{N_1}})(x) = F_X^{-1}(F_X^{\frac{N_2}{N_1}\frac{n_1}{n_2}})(x).
$$

□

It is interesting to note that the paradox probability $\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) = \frac{n_2}{n_1+n_2}$ as $N_1 \to \infty$ under the setup of Lemma 3.2.

We verify Lemma 3.2 with a Monte Carlo simulation. In each simulation, we first randomly generate two groups of observation: Y_1 and Y_2 from the standard normal distribution, and the number of each group is given by n_1 and n_2. Based on our setting, we assume the latent aptitudes follow the Pareto distribution with $\alpha = 1.16$. Then we obtain the latent aptitudes for each group and check whether the paradox happens. We repeat this process 1000 times and record the frequency of the paradox. 1000 Monte Carlo iterations of our simulation are performed and we compute the mean of the frequency.

	$n_1 = 400, n_2 = 600, N_1 = 1e6, N_2 = 1e6$	$n_1 = 300, n_2 = 700, N_1 = 1e6, N_2 = 1e6$
Prob from Lemma 3.2	0.1	0.2
Prob from simulation	0.10084	0.200309

Table 1: Simulation result to validate the lemma about the probability of paradox

Lemma 3.3. *Suppose that the system inclusion mechanism has the property that* $g(X_{i,(N_i)}) = 1$*, and that the Full House Model is of the form*

$$\begin{aligned} Y_{i,j} &\sim F_Y(y), \qquad i = 1, 2, \qquad j = 1, \ldots, n_1, \\ X_{i,j} &\sim F_X(x), \qquad i = 1, 2, \qquad j = 1, \ldots, N_i, \end{aligned}$$

where $F_Y^{-1}(y)$ *and* $F_X^{-1}(x)$ *exist. Then,*

$$\mathrm{P}\left(X_{1,(1)} > X_{2,(1)}, Y_{1,(1)} < Y_{2,(1)}\right) = \frac{N_2}{N_1 + N_2} - \frac{n_2}{n_1 + n_2}.$$

Proof. We can use the convenient representation of (7) since the conditions of Proposition 3.1 hold. Thus, we have

$$\begin{aligned} &\mathrm{P}\left(X_{1,(1)} > X_{2,(1)}, Y_{1,(1)} < Y_{2,(1)}\right) = \mathrm{P}\left(X_{2,(1)} < X_{1,(1)} < h_{1,2}\left(X_{2,(1)}\right)\right) \\ &= \int_{-\infty}^{+\infty} \left[\int_x^{h_{1,2}(x)} f_{X_{1,(1)}}(y)\, dy\right] f_{X_{2,(1)}}(x)\, dx \\ &= \int_{-\infty}^{+\infty} \left[\int_x^{h_{1,2}(x)} N_1(1 - F_X(y))^{N_1 - 1}\, f_X(y) dy\right] N_2(1 - F_X(x))^{N_2 - 1}\, f_X(x)\, dx \\ &= \int_{-\infty}^{+\infty} \left[\int_x^{F_X^{-1}(1-(1-F_X(x))^{\frac{N_2}{N_1}\frac{n_1}{n_2}})} N_1(1 - F_X(y))^{N_1 - 1}\, f_X(y) dy\right] N_2(1 - F_X(x))^{N_2 - 1}\, f_X(x)\, dx \\ &= \int_{-\infty}^{+\infty} -((1 - F_X(x))^{N_2 \frac{n_1}{n_2}} + (1 - F_X(x))^{N_1})\, N_2(1 - F_X(x))^{N_2 - 1}\, f_X(x)\, dx \\ &= \int_0^1 -(1 - t)^{\frac{n_1}{n_2}} + (1 - t)^{\frac{N_1}{N_2}}\, dt \\ &= \frac{N_2}{N_1 + N_2} - \frac{n_2}{n_1 + n_2}, \end{aligned}$$

where

$$h_{1,2}(x) = F_X^{-1}(1 - (F_Y(F_Y^{-1}((1 - F_X)^{\frac{N_2}{n_2}})))^{\frac{n_1}{N_1}})(x) = F_X^{-1}(1 - (1 - F_X)^{\frac{N_2}{N_1}\frac{n_1}{n_2}})(x).$$

□

Lemma 3.4. *If we assume*

$$\begin{aligned} Y_{i,j} &\sim F_Y(y), \qquad i = 1, 2, \qquad j = 1, \ldots, n_1, \\ X_{i,j} &\sim F_X(x), \qquad i = 1, 2, \qquad j = 1, \ldots, N_i, \end{aligned}$$

where $F_Y^{-1}(y)$ *and* $F_X^{-1}(x)$ *are assumed to exist. Then,*

$$\mathrm{P}\left(X_{1,(N_1-k)} > X_{2,(N_2-j)}, Y_{1,(n_1-k)} < Y_{2,(n_2-j)}\right) = .$$

Proof.

$$\begin{aligned}
&\mathrm{P}\left(X_{1,(N_1-k)} > X_{2,(N_2-j)}, Y_{1,(n_1-k)} < Y_{2,(n_2-j)}\right) = \mathrm{P}\left(X_{2,(N_2-j)} < X_{1,(N_1-k)} < h_{1,2}\left(X_{2,(N_2-j)}\right)\right) \\
&= \int_{-\infty}^{+\infty} \left[\int_x^{h_{1,2}(x)} f_{X_{1,(N_1-k)}}(y)\, dy\right] f_{X_{2,(N_2-j)}}(x)\, dx \\
&= \int_{-\infty}^{+\infty} \left[\int_x^{h_{1,2}(x)} \frac{N_1!}{(N_1-k-1)!\, k!}(F_X(y))^{N_1-k-1}(1-F_X(y))^k\, f_X(y)dy\right] f_{X_{2,(N_2-j)}}(x)\, dx \\
&= \int_{-\infty}^{+\infty} \sum_{t=N_1-k}^{N_1} \binom{N_1}{j} \left((F_X(h_{1,2}(x)))^j(1-F_X(h_{1,2}(x)))^{N_1-j} - (F_X(x))^j(1-F_X(x))^{N_1-j}\right) dy \\
&f_{X_{2,(N_2-j)}}(x)\, dx
\end{aligned}$$

where

$$\begin{aligned}
&h_{1,2}(x) = (g_1 \circ f_2)(x) \\
&= F^{-1}\left(I^{-1}(N_1-k, k+1)I(n_1-k, k+1)I^{-1}(n_2-j, j+1)I_{F_X(X_{2,(N_2-j)})}(N_2-j, j+1)\right)
\end{aligned}$$

and $I_x(\alpha, \beta)$ is the regularized incompleted beta function. □

Definition 3.5. *We will say that Y_2 is first order ε-weakly stochastically dominant over Y_1 if, for $\varepsilon > 0$, $\mathrm{P}(Y_2 \le y) \le \mathrm{P}(Y_1 \le y) \le \mathrm{P}(Y_2 \le y) + \varepsilon$ for all y and $\mathrm{P}(Y_2 \le y) < \mathrm{P}(Y_1 \le y) \le \mathrm{P}(Y_2 \le y) + \varepsilon$ for some y.*

Lemma 3.6. *Suppose that the system inclusion mechanism has the property that $g(X_{i,(N_i)}, \boldsymbol{X}_{i,-(N_i)}) = 1$, and that the Full House Model is of the form*

$$\begin{aligned}
Y_{i,j} &\sim F_{Y_i}(y), \qquad i = 1, 2, \qquad j = 1, \ldots, n_i, \\
X_{i,j} &\sim F_X(x), \qquad i = 1, 2, \qquad j = 1, \ldots, N_i,
\end{aligned}$$

where $F_{Y_i}^{-1}(y)$ and $F_X^{-1}(x)$ exist. Further suppose that Y_2 is first order ε-weakly stochastically dominant over Y_1 for some $\varepsilon > 0$. Then,

$$\begin{aligned}
\frac{n_2}{n_1+n_2} - \frac{N_2}{N_1+N_2} &\le \mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) \\
&\le n_2 \sum_{i=0}^{n_1} \frac{\binom{n_1}{i}\varepsilon^i}{n_1+n_2-i} - \frac{N_2}{N_1+N_2}
\end{aligned} \tag{8}$$

Proof. First, observe that the assumption of first order ε-weakly stochastically dominant gives

$$\left(F_{Y_{2,(n_2)}}(y)\right)^{n_1/n_2} \le (F_{Y_1}(y))^{n_1} \le (F_{Y_2}(y)+\varepsilon)^{n_1} = \left(\left(F_{Y_{2,(n_2)}}(y)\right)^{1/n_2} + \varepsilon\right)^{n_1},$$

and this implies that

$$F_X^{-1}\left(F_X^{\frac{N_2}{N_1}\frac{n_1}{n_2}}(X_{2,(N_2)})\right) \leq h_{1,2}(X_{2,(N_2)}) \leq F_{X_{1,(N_1)}}^{-1}\left(\left(\left(F_{X_{2,(N_2)}}(X_{2,(N_2)})\right)^{1/n_2}+\varepsilon\right)^{n_1}\right).$$

In combination with Lemma 3.2 and the proof of Lemma 3.2, the first inequality yields

$$\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) \geq \frac{n_2}{n_1+n_2} - \frac{N_2}{N_1+N_2}.$$

This verifies the first inequality of (8). We now verify the second inequality of (8). Now,

$$\begin{aligned}
&\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) \\
&\quad\leq \int_{-\infty}^{\infty}\left[\int_x^{F_{X_{1,(N_1)}}^{-1}\left(\left(\left(F_{X_{2,(N_2)}}(x)\right)^{1/n_2}+\varepsilon\right)^{n_1}\right)} f_{X_{1,(N_1)}}(y)dy\right] f_{X_{2,(N_2)}}(x)dx \\
&\quad= \int_{-\infty}^{\infty}\left[\left((F_X(x))^{N_2/n_2}+\varepsilon\right)^{n_1} - F_X(x)^{N_1}\right] N_2 F_X(x)^{N_2-1} f_X(x)dx \\
&\quad= \int_{-\infty}^{\infty}\left((F_X(x))^{N_2/n_2}+\varepsilon\right)^{n_1} N_2 F_X(x)^{N_2-1} f_X(x)dx - \frac{N_2}{N_1+N_2} \\
&\quad= \int_0^1 \left(t^{1/n_2}+\varepsilon\right)^{n_1} dt - \frac{N_2}{N_1+N_2} \\
&\quad= n_2 \sum_{i=0}^{n_1} \frac{\binom{n_1}{i}\varepsilon^i}{n_1+n_2-i} - \frac{N_2}{N_1+N_2}
\end{aligned}$$

□

Example: Uniform distributions. In this example we demonstrate a case where Lemma 3.6 applies, but we can explicitly calculate the paradox probability. Suppose that the system inclusion mechanism has the property that $g(X_{i,(N_i)}, \mathbf{X}_{i,-(N_i)}) = 1$, and we will consider the following Full House Model for two systems,

$$\begin{aligned}
Y_{1,j} &\sim U(0,1), \qquad j = 1,\ldots,n_1, \\
Y_{2,j} &\sim U(0,1+\delta), \qquad j = 1,\ldots,n_2, \\
X_{i,j} &\sim F_X(x), \qquad j = 1,\ldots,N_i,
\end{aligned}$$

where $\delta > 0$. We can pick $\varepsilon = \delta/(1+\delta)$ so that Y_2 is first order ε-weakly stochastically dominant over Y_1 with this choice of ε, and the conditions of Lemma 3.6 are satisfied. However, our setup allows for direct calculation of the paradox probability. From Proposition 3.1 and

a bit of manipulation we have that

$$
\begin{aligned}
&\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) = \mathrm{P}\left(X_{2,(N_2)} < X_{1,(N_1)} < h_{1,2}\left(X_{2,(N_2)}\right)\right) \\
&= \int_{-\infty}^{+\infty} \int_{x}^{h_{1,2}(x)} f_{X_{1,(N_1)}}(y)\ dy\ f_{X_{2,(N_2)}}(x)\ dx \\
&= \int_{-\infty}^{+\infty} \int_{x}^{h_{1,2}(x)} N_1 F_X(y)^{N_1-1}\ dy\ N_2 F_X(x)^{N_2-1}\ dx \\
&= \int_{-\infty}^{+\infty} \int_{x}^{F_X^{-1}(((1+\delta)F_X^{\frac{N_2}{n_2}})^{\frac{n_1}{N_1}})(x)} N_1 F_X(y)^{N_1-1}\ dy\ N_2 F_X(x)^{N_2-1}\ dx \\
&= \int_{-\infty}^{+\infty} (((1+\delta)F_X(x)^{\frac{N_2}{n_2}})^{n_1} - F_X(x)^{N_1})\ N_2 F_X(x)^{N_2-1}\ dx \\
&= \int_{0}^{1} (1+\delta)^{n_1} t^{\frac{n_1}{n_2}} - t^{\frac{N_1}{N_2}}\ dt \\
&= \frac{n_2}{n_1+n_2}(1+\delta)^{n_1} - \frac{N_2}{N_1+N_2}.
\end{aligned}
$$

Here we see that the probability of paradox is 1 for all $\delta \geq [2(n_1/n_2+1)]^{1/n_1} - 1$. When $n_1 \gg n_2$, then $[2(n_1/n_2+1)]^{1/n_1} - 1 \approx n_1^{n_1} - 1$ and we see that the probability of paradox is 1 for extremely small values of δ.

Example: Normal distributions

In this example, we demonstrate a case where Lemma 3.6 applies. Suppose that the system inclusion mechanism has the property that $g(X_{i,(N_i)}, \mathbf{X}_{i,-(N_i)}) = 1$, and we will consider the following Full House Model for two systems,

$$
\begin{aligned}
Y_{1,j} &\sim N(\mu_1, \sigma_1^2), \qquad j = 1, \ldots, n_1, \\
Y_{2,j} &\sim N(\mu_2, \sigma_2^2), \qquad j = 1, \ldots, n_2, \\
X_{i,j} &\sim F_X(x), \qquad j = 1, \ldots, N_i,
\end{aligned}
$$

where $\mu_1 \leq \mu_2, \sigma_1 > 0$ and $\sigma_2 > 0$.

Supposed $\sigma_1 = \sigma_2 = \sigma$, it is obvious to see $\mathrm{P}(Y_2 \leq y) \leq \mathrm{P}(Y_1 \leq y)$ and we can also find the ε such that $\mathrm{P}(Y_1 \leq y) \leq \mathrm{P}(Y_2 \leq y) + \varepsilon$

$$
\begin{aligned}
F_{Y_1}(y) - F_{Y_2}(y) &= \int_{-\infty}^{y} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu_1)^2}{2\sigma^2}} dx - \int_{-\infty}^{y} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu_2)^2}{2\sigma^2}} dx \\
&= \int_{y-\frac{\mu_2-\mu_1}{2}}^{y+\frac{\mu_2-\mu_1}{2}} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\frac{\mu_2+\mu_1}{2})^2}{2\sigma^2}} dx
\end{aligned}
$$

Since $\frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\frac{\mu_2+\mu_1}{2})^2}{2\sigma^2}}$ is the density function for $N(\frac{\mu_2+\mu_1}{2}, \sigma^2)$, then the maximum value of the difference between these two cumulative distribution function of the normal distribution is achieved when $y = \frac{\mu_2+\mu_1}{2}$.

We can pick $\varepsilon = \int_{\mu_1}^{\mu_2} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\frac{\mu_2+\mu_1}{2})^2}{2\sigma^2}} dx$ so that Y_2 is first order ε-weakly stochastically dominant over Y_1 with this choice of ε, and the conditions of Lemma 3.6 are satisfied.

From Proposition 3.1 and a bit of manipulation, we have that

$$
\begin{aligned}
&\mathrm{P}\left(X_{1,(N_1)} > X_{2,(N_2)}, Y_{1,(n_1)} < Y_{2,(n_2)}\right) = \mathrm{P}\left(X_{2,(N_2)} < X_{1,(N_1)} < h_{1,2}\left(X_{2,(N_2)}\right)\right) \\
&= \int_{-\infty}^{+\infty} \int_{x}^{h_{1,2}(x)} f_{X_{1,(N_1)}}(y)\ dy\ f_{X_{2,(N_2)}}(x)\ dx \\
&= \int_{-\infty}^{+\infty} \int_{x}^{h_{1,2}(x)} N_1 (F_X(y))^{N_1-1}\ dy\ N_2 F_X(x)^{N_2-1}\ dx \\
&= \int_{-\infty}^{+\infty} \int_{x}^{F_X^{-1}((\Phi(\Phi^{-1}(F_X^{\frac{N_2}{n_2}})+\frac{\mu_2-\mu_1}{\sigma}))^{\frac{n_1}{N_1}})(x)} N_1 F_X(y)^{N_1-1}\ dy\ N_2 F_X(x)^{N_2-1}\ dx \\
&= \int_{-\infty}^{+\infty} ((\Phi(\Phi^{-1}(F_X(x)^{\frac{N_2}{n_2}}) + \frac{\mu_2-\mu_1}{\sigma}))^{n_1} - F_X(x)^{N_1})\ N_2 F_X(x)^{N_2-1}\ dx \\
&= \int_{0}^{1} (\Phi(\Phi^{-1}(t^{\frac{1}{n_2}}) + \frac{\mu_2-\mu_1}{\sigma}))^{n_1} - t^{\frac{N_1}{N_2}})\ dt \\
&= \int_{0}^{1} (\Phi(\Phi^{-1}(t^{\frac{1}{n_2}}) + \frac{\mu_2-\mu_1}{\sigma}))^{n_1}\ dt - \frac{N_2}{N_1+N_2} \\
&= \int_{-\infty}^{+\infty} \Phi(x+\frac{\mu_2-\mu_1}{\sigma})^{n_1} (n_2 \Phi(x)^{n_2-1}\phi(x))\ dx - \frac{N_2}{N_1+N_2} \\
&= n_2 \int_{-\infty}^{+\infty} \mathbb{P}(Z_{1,1} - X \le \frac{\mu_2-\mu_1}{\sigma}, \ldots, Z_{1,n_1} - X \le \frac{\mu_2-\mu_1}{\sigma}, Z_{2,1} - X \le 0, \ldots, Z_{2,n_2-1} - X \le 0 | X = x)\phi(x \\
&- \frac{N_2}{N_1+N_2} \\
&= n_2 \mathbb{P}(Z_{1,1} - X \le \frac{\mu_2-\mu_1}{\sigma}, \ldots, Z_{1,n_1} - X \le \frac{\mu_2-\mu_1}{\sigma}, Z_{2,1} - X \le 0, \ldots, Z_{2,n_2-1} - X \le 0) - \frac{N_2}{N_1+N_2} \\
&= n_2 F_{\mathbf{W}}(\mathbf{w} = (\frac{\mu_2-\mu_1}{\sigma}, \ldots, \frac{\mu_2-\mu_1}{\sigma}, 0, \ldots, 0)) - \frac{N_2}{N_1+N_2}
\end{aligned}
$$

where

$$
\begin{aligned}
h_{1,2}(x) &= (g_1 \circ f_2)(x) \\
&= F_X^{-1}((F_{Y_2}(F_{Y_1}^{-1}(F_X^{\frac{N_2}{n_2}})))^{\frac{n_1}{N_1}})(x) \\
&= F_X^{-1}((\Phi(\Phi^{-1}(F_X^{\frac{N_2}{n_2}}) - \frac{\mu}{\sigma}))^{\frac{n_1}{N_1}})(x)
\end{aligned}
$$

$\mathbf{W} \sim \mathcal{N}_{(n_1+n_2-1)}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with (n_1+n_2-1)-dimensional mean vector

$$\boldsymbol{\mu} = (0, \ldots, 0)$$

and $(n_1+n_2-1)\times(n_1+n_2-1)$ covariance matrix

$$\mathbf{\Sigma}=\begin{bmatrix}2 & 1 & \dots & 1\\ 1 & 2 & \ddots & \vdots\\ \vdots & \ddots & \ddots & 1\\ 1 & \dots & 1 & 2\end{bmatrix}$$

The first n_1 elements of $\mathbf{w}$ is $\frac{\mu_2-\mu_1}{\sigma}$ and the last n_2-1 elements are 0.

Then when $\sigma_1\neq\sigma_2$, we do not have a closed form for ϵ but we can calculate the probability of the paradox.

$$\begin{aligned}
&\mathrm{P}\left(X_{1,(N_1)}>X_{2,(N_2)},Y_{1,(n_1)}<Y_{2,(n_2)}\right)=\mathrm{P}\left(X_{2,(N_2)}<X_{1,(N_1)}<h_{1,2}\left(X_{2,(N_2)}\right)\right)\\
&=\int_{-\infty}^{+\infty}\int_{x}^{h_{1,2}(x)} f_{X_{1,(N_1)}}(y)\ dy\ f_{X_{2,(N_2)}}(x)\ dx\\
&=\int_{-\infty}^{+\infty}\int_{x}^{h_{1,2}(x)} N_1(F_X(y))^{N_1-1}\ dy\ N_2F_X(x)^{N_2-1}\ dx\\
&=\int_{-\infty}^{+\infty}\int_{x}^{F_X^{-1}(\Phi(\frac{\Phi^{-1}(F_X^{\frac{N_2}{n_2}})\sigma_2+\mu_2-\mu_1}{\sigma_1})^{\frac{n_1}{N_1}})(x)} N_1F_X(y)^{N_1-1}\ dy\ N_2F_X(x)^{N_2-1}\ dx\\
&=\int_{-\infty}^{+\infty}((\Phi(\frac{(\Phi^{-1}(F_X(x)^{\frac{N_2}{n_2}})\sigma_2+\mu_2-\mu_1}{\sigma_1}))^{n_1}-F_X(x)^{N_1})\ N_2F_X(x)^{N_2-1}\ dx\\
&=\int_0^1(\Phi(\frac{\Phi^{-1}(t^{\frac{1}{n_2}})\sigma_2+\mu_2-\mu_1}{\sigma_1}))^{n_1}-t^{\frac{N_1}{N_2}})\ dt\\
&=\int_0^1(\Phi(\frac{\Phi^{-1}(t^{\frac{1}{n_2}})\sigma_2+\mu_2-\mu_1}{\sigma_1}))^{n_1}\ dt-\frac{N_2}{N_1+N_2}\\
&=\int_{-\infty}^{+\infty}\Phi(\frac{\sigma_2x+\mu_2-\mu_1}{\sigma_1})^{n_1}(n_2\Phi(x)^{n_2-1}\phi(x))\ dx-\frac{N_2}{N_1+N_2}\\
&=n_2\int_{-\infty}^{+\infty}\mathbb{P}(Z_{1,1}-\frac{\sigma_2}{\sigma_1}X\leq\frac{\mu_2-\mu_1}{\sigma_1},\dots,Z_{1,n_1}-\frac{\sigma_2}{\sigma_1}X\leq-\frac{\mu_2-\mu_1}{\sigma_1},\\
&Z_{2,1}-X\leq0,\dots,Z_{2,n_2-1}-X\leq0|X=x)\phi(x)\ dx-\frac{N_2}{N_1+N_2}\\
&=n_2\mathbb{P}(Z_{1,1}-\frac{\sigma_2}{\sigma_1}X\leq\frac{\mu_2-\mu_1}{\sigma_1},\dots,Z_{1,n_1}-\frac{\sigma_2}{\sigma_1}X\leq-\frac{\mu_2-\mu_1}{\sigma_1},Z_{2,1}-X\leq0,\dots,Z_{2,n_2-1}-X\leq0)-\frac{N_2}{N_1+}\\
&=n_2F_{\mathbf{W}}(\mathbf{w}=(\frac{\mu_2-\mu_1}{\sigma_1},\dots,\frac{\mu_2-\mu_1}{\sigma_1},0,\dots,0))-\frac{N_2}{N_1+N_2}
\end{aligned}$$

where

$$
\begin{aligned}
h_{1,2}(x) &= (g_1 \circ f_2)(x) \\
&= F_X^{-1}((F_{Y_2}(F_{Y_1}^{-1}(F_X^{\frac{N_2}{n_2}})))^{\frac{n_1}{N_1}})(x) \\
&= F_X^{-1}((\Phi(\frac{\Phi^{-1}(F_X^{\frac{N_2}{n_2}})\sigma_2 + \mu_2 - \mu_1}{\sigma_1}))^{\frac{n_1}{N_1}})(x)
\end{aligned}
$$

$\mathbf{W} \sim \mathcal{N}_{(n_1+n_2-1)}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with $(n_1 + n_2 - 1)$-dimensional mean vector

$$\boldsymbol{\mu} = (0, \ldots, 0)$$

and $(n_1 + n_2 - 1) \times (n_1 + n_2 - 1)$ covariance matrix

$$\boldsymbol{\Sigma} = \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix}$$

$$\Sigma_{11} = \begin{bmatrix} \frac{\sigma_2^2}{\sigma_1^2} + 1 & \frac{\sigma_2^2}{\sigma_1^2} & \ldots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \frac{\sigma_2^2}{\sigma_1^2} \\ \frac{\sigma_2^2}{\sigma_1^2} & \ldots & \frac{\sigma_2^2}{\sigma_1^2} & \frac{\sigma_2^2}{\sigma_1^2} + 1 \end{bmatrix}, \Sigma_{12} = \Sigma_{21} = \begin{bmatrix} \frac{\sigma_2}{\sigma_1} & \ldots & \ldots & \frac{\sigma_2}{\sigma_1} \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\sigma_2}{\sigma_1} & \ldots & \ldots & \frac{\sigma_2}{\sigma_1} \end{bmatrix}, \Sigma_{22} = \begin{bmatrix} 2 & 1 & \ldots & 1 \\ 1 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ 1 & \ldots & 1 & 2 \end{bmatrix}$$

The first n_1 elements of $\mathbf{w}$ is $\frac{\mu_2 - \mu_1}{\sigma_1}$ and the last $n_2 - 1$ elements are 0.

4 Educational Attainment Simulation

Our Gould paradox is validated by simulating the process of students moving on to the next level of education. First, we randomly generate 1 million talents from Pareto distribution, which correspond to the 1 million talents of primary school students. Then we randomly generate 2 thousand school indices for these 1 million talents, which represent the schools that the students will attend. Finally, we transform ordered talents corresponding to the eligible population to percentiles from the ordered statistics and convert percentiles to order scores from a normal distribution. The score distribution for each school is represented by the normal distribution. The mean of the Normal distribution ranges from 60 to 80 and the standard deviation is 10, which indicates the different levels of schools. Better schools have higher mean scores. When compared to other students, we wonder if the most talented primary student has lower scores with a larger mean of normal distribution. In other words, we wonder if the most talented primary student, who attends a better school, still underperforms the students who attend schools with lower mean scores.

100 Monte Carlo iterations of our simulation are performed. The results are depicted in Table 2. Based on the results in Table 2, the most talented students could perform worse than the other students who attend schools with lower mean scores.

num of talent paradox happen	average of μ of dist that generate top 10 scores
3435	79.38269
2477	79.62781
3386	78.96648
37	78.26713
1995	79.67984
3498	79.59680
74852	78.81841
5272	79.00650
970	79.41571
302	79.58979

Table 2: The first column shows the number of primary students who have higher scores than the most talented primary students with a lower mean of the normal distribution. The second column displays the average means of normal distributions that generate top 10 scores.

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